SMART HOME APPLIANCES AND NOTIFICATION SYSTEMS

ABSTRACT

This report serves as the documentation of our semester final project for the EEE1002 (Introduction to Electrics & Electronics Engineering) course. It offers an insightful overview of the application of electrics and electronics in smart home appliances and notification systems. The project revolves around the implementation of electronics for an ideally integrated smart home system, controllable via mobile application(s).

The components utilized in this project include Arduino Mega 2560, ESP32, transistors, diodes, and resistors. Initially, we conceptualized the design through sketches on paper, followed by assembling the model using wooden parts. Subsequently, we developed the necessary code for the accompanying application.

The VCC is connected to the collector pin, while a 1k ohm resistor is connected to the base, which in turn is connected to the Arduino; it closes the circuit when electricity is applied. The emitter is connected to LEDs, motors and buzzers which are connected to ground that completes the circuit. This structure, utilizing principles of electricity and electronics, forms a basic control mechanism of our smart home system. Arduino establishes connections among these components, enabling efficient management of our smart home system through the application. The reason why we used transistors is that the Arduino cannot effectively put enough current to power all of the connected devices concurrently. Arduino serves as the circuit controller. We employed the ESP32 (Wi-Fi module) to establish a connection with the server hosting the control mechanism. Communication between the Arduino and ESP32 occurs through a serial connection. Application that controls the ESP32 and Arduino is coded with C++. The server is coded with javascript.

Table of Contents

Chapter 1

1.1 Purpose of the Project

The central focus of this project revolves around the integration of electronic systems to enable the synchronized, and automated control of various essential functionalities within a household setting.

These functionalities span across a multitude of systems, encompassing tasks such as the regular retrieval and integration of weather updates into the household network, management of lighting systems, control of curtains for privacy, and light regulation, the nuanced adjustment of ceiling fan for comfort, the implementation of dynamic day-night routines and security measures through the integration of keypad.

The overarching objective of this endeavor is to electrify these disparate systems and establish robust, interconnected communication channels between them. By seamlessly interlinking these functionalities, the project seeks to showcase a comprehensive and cohesive Internet of Things (IoT) framework within the domestic sphere, illustrating the potential for enhanced convenience, efficiency, and comfort through the convergence of electronic automation and intelligent networked systems.

The project served as a valuable learning opportunity for our team, offering a view into automation systems, circuit design, and collaborative teamwork. Our primary objective was not only to achieve the project goals but also to broaden our understanding in these domains.

1.2 Reason of Choosing This Project

Automated home systems are a growing market and an important aspect of the general trend towards integrating smart technologies into daily life. By exploring this topic, we aim to address a relevant and timely area of interest.

The field of automated home systems is constantly evolving, with new advances and technologies emerging on a regular basis. By addressing this topic, we can explore innovative solutions and contribute to the ongoing dialogue about smart home technology.

Automated home systems offer physical benefits to home-owners, including increased comfort, energy efficiency and security. By developing a project in this area, we can directly address real needs and demonstrate the practical applications of new technologies.

Researching automated home systems is a valuable learning opportunity for our team, allowing us to gain hands-on experience with cutting-edge technologies such as IoT devices, machine learning algorithms and home automation platforms. This project will enhance our skills and knowledge in relevant areas of technology and engineering.

Our decision to choose automated home systems as the focus of our project is based on their relevance, innovation potential, practical application and learning opportunities.

1.3 Applications on the Project

- **Periodical Weather Updates**

  Our system has been meticulously crafted to seamlessly integrate with a Weather API, allowing for the retrieval of hourly weather data. This data is then displayed on an LCD screen, providing residents with up-to-the-hour updates on weather conditions outside their home. By leveraging this capability, occupants can make informed decisions about their daily activities, plan outdoor ventures accordingly, and optimize indoor comfort settings in response to changing weather patterns.

- **Lighting Systems**

  Our lighting systems have been designed to offer residents control and flexibility. Utilizing remote devices, occupants can effortlessly adjust the LED lights at their convenience.

- **Curtains Control**

  Our curtain control system combines the convenience of both automated day-night routines and manual control via remote devices. By implementing a day-night routines protocol, residents can effortlessly automate the opening and closing of curtains according to predefined schedules, and for privacy throughout the day. Additionally, occupants can override these automated routines and manually adjust the curtains using remote devices.

- **Ceiling Fan Control**

  The ceiling fan system has been designed to offer seamless control via remote devices. This control mechanism allows occupants to tailor their indoor environment to their liking, enhancing comfort and convenience within the home.

- **Day-Night Routines**

  The Day-Night Routines protocol serves as a framework within our system, that dynamically activates or deactivates integrated systems based on time of day. By leveraging this protocol, our system can automatically adjust various functionalities such as lighting, curtain operation, and other systems according to predefined schedules.

- **Security**

  The security system ensures exclusive control over integrated systems by residents. Utilizing a keypad and a unique code known only to residents, it grants access for system activation, providing a secure and personalized environment within the household.

1.4 Controlling the System from a Remote Device

In order to control the whole system with ease we created a server that runs on an external device. Server runs on the basic http protocol and communicates with user and the system at the same time. While giving access to the user for controlling the system like turning on/off the lights or moving curtains, the server can also provide data to the system like weather info.

With the limitations of ESP32 we had to use polling in order to check if the user decided to change the state of any device. ESP32 periodically sends requests to the server with the defined rate of 500ms. Because delaying the main loop would block the code from continuing and prevent other functionalities of the system from running, we implemented a timer system that updates multiple timers at once and only run the function that associated with the timer once the timeout has been reached. Therefore, the system can fetch device updates, weather, date and clock without blocking.
The timer is implemented as an object that stores time that has passed since the last call to the associated function, a timeout, and route. Timers are updated from the main loop every time it reaches to the end of the loop with the millis() function provided from the Arduino library. When a timer reaches the timeout value associated function is called and timer resets. In this way we achieved asynchronous like functionality without needing to wait on multiple delay() functions.

- **Communication Protocol**

On top of the http protocol system and server uses a simple protocol that makes the communication easier to happen. A client sends a '/changeDeviceState' request to the server with the device id whose state should be changed. Then the server adds it to an array of ids that should be changed. When esp32's timer for device updates reaches to end it sends a '/pollDeviceUpdates' to the server. Then the server creates a message header that contains message type, number of updates, two empty bytes and then fills the remaining buffer with updated devices as bytes. So, the header consists of four bytes packed into an unsigned integer. Therefore, the size of the message is reduced. Other than the message header the three bytes can be used as arguments depending on the message type as it is used for storing number of updates in the device update message.

Chapter 2

2.1 Usage of Components

- **Arduino Mega 2560** is the central controller in this circuit. It interfaces with the other components as follows:

  Keypad: The keypad is connected to digital pins 37, 36, 35, and 34 (columns) and 33, 32, 31, and 30 (rows). The Arduino reads the keypad inputs to detect button presses.

  Transistor Stack: The Arduino controls the transistors connected to digital pins 22 through 29 and 39 through 46. These transistors are used to drive the LEDs.

  LCD Screen: The LCD screen is connected to digital pins 49 through 53. These pins are used for data (DB4-DB7) and control signals (RS, RW, E).

- **ESP32** is connected to the Arduino Mega 2560 via the serial interface (RX/TX):

  RX/TX: The RX and TX pins on the Arduino (pins 0 and 1) are connected to the TX and RX pins of the ESP32. This setup allows for serial communication between the two microcontrollers.

- **Transistors**

  2N2222A Transistors: These are NPN transistors that act as switches. They are controlled by the Arduino through the base resistors (R1 to R10).

  Base Resistors: Each transistor has a base resistor (770Ω) that limits the current to the base, ensuring the transistors switch properly.

  Outputs: The collectors of the transistors are connected to the LEDs, and the emitters are connected to ground. When the Arduino outputs a high signal, it turns on the corresponding transistor, allowing current to flow through the LED and light it up.

- **Resistors**

  The resistors are used in several places in the circuit:

  Base Resistors (R1 to R10): These 770Ω resistors limit the current to the bases of the transistors.

  LED Current Limiting Resistors (R11 to R18): These 110Ω resistors limit the current through the LEDs to prevent them from burning out.

  Pull-Down Resistors (R19 to R21): These resistors ensure that the LCD screen's control pins are pulled to ground when not actively driven by the Arduino, preventing floating inputs.

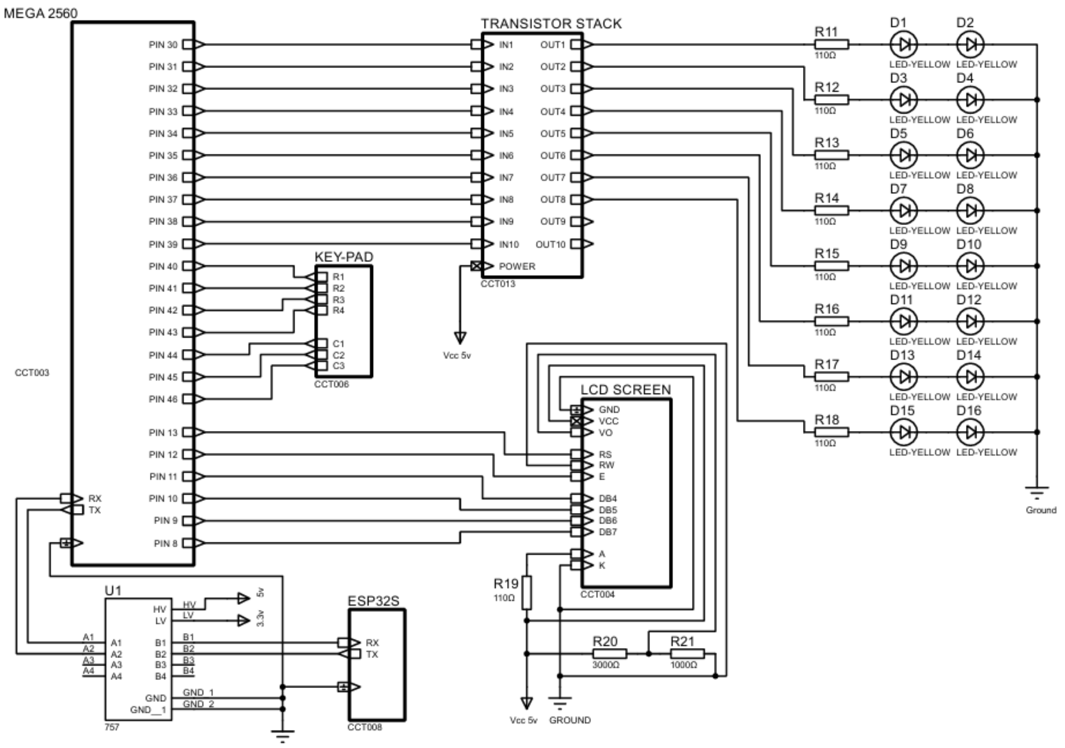
- **Diodes**

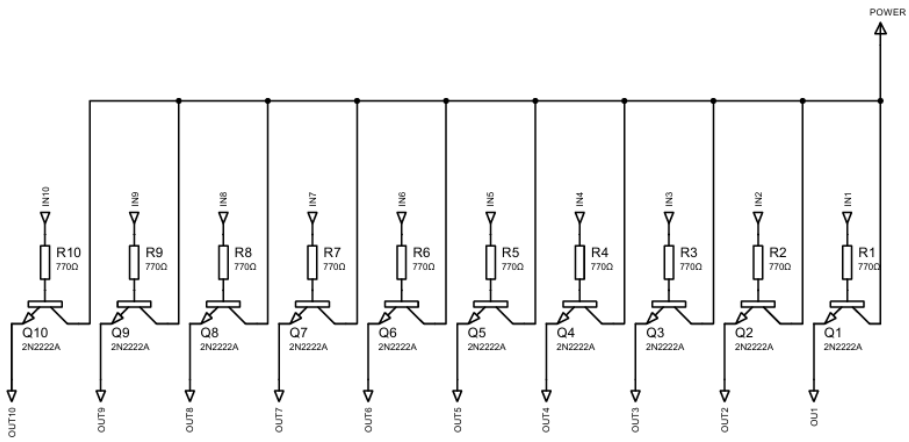
  The diodes in this circuit are LEDs (D1 to D16) connected to the transistor stack:

LEDs (D1 to D16): These yellow LEDs are controlled by the transistors. Each LED is connected in series with a current-limiting resistor (R11 to R18).

Chapter 3

3.1 Schematics





3.2 Evaluation and Achievements

During project development, our team has acquired skills and experience that have contributed significantly to our success. Here you will find a comprehensive evaluation of our services:

- ## Practical skills:

Circuit design and analysis: We have familiarized ourselves with designing circuits, understanding resistance values and calculating limits for components such as LEDs and transistors to ensure optimal performance and safety.

Power supply design: Through careful calculations, we determined the power requirements of our devices and designed appropriate power supplies to meet those needs and improve the reliability and efficiency of our system.

Serial communication: We familiarized ourselves with serial communication protocols that enable data transfer between the various components of our automated home system.

CAD and CNC knowledge: Using CAD software, we efficiently designed and prepared files for CNC machining to ensure precision and accuracy in manufacturing the physical components for our project.

- ## Technical Skills:

Website design: We developed a user-friendly website for our project.

HTTP protocols: By understanding HTTP protocols, we were able to implement effective communication between our automated home system and external devices.

Cable Management: Implementing effective cable management techniques kept our project neat, tidy and organized, reduced clutter and minimized the risk of electrical hazards.

- ## Project management and leadership:

Effective time management: We optimized our workflow and maximized productivity by efficiently allocating time to various project tasks, ensuring timely completion and delivery.

Distribution of issues: By assigning tasks based on team members' strengths and expertise, we effectively distributed the workload and tackled challenges in a systematic way, which fostered collaboration.

Effective communication: Clear and concise communication facilitated coordination between team members and ensured that everyone was aligned with the project goals.

- ## Safety and compliance:

Electrical safety: While developing our project, we took great care to ensure that LEDs and transistors were not damaged. Through careful circuit design and careful calculations of resistance values, we ensured that the components were operated within safe operating limits. This proactive approach minimized the risk of overload and protected the integrity of our electronic components, contributing to the reliability and longevity of our automated home system

Overall, on our journey to develop the automated home system, we not only gained valuable technical skills, but also learned the importance of effective project management, teamwork and safety compliance. Through our combined efforts and commitment, we have successfully realized our project goals and emerged with a deeper understanding of the complexities and intricacies of integrating modern technologies into everyday life.

3.3 Shortcomings & Difficulties

- ### Technical Challenges

Code Integration: The foremost technical challenge was integrating the codes for the ESP32 and Arduino. Initially, we had separate codes for each device, and combining them to function cohesively posed a significant difficulty. Nonetheless, we eventually managed to integrate the codes successfully.

Tool Proficiency: Another technical obstacle was our limited experience with the necessary tools. However, we perceived this as an opportunity to gain valuable hands-on experience, which we gradually acquired throughout the project.

Wiring and Cable Management: We also struggled with organizing the wiring and managing the cables. Despite the model being sufficiently large to house all the cables, we needed to arrange them efficiently to maximize space and improve the overall appearance. Thankfully, we were able to address this issue promptly.

- ### Materialistic Challenges

We faced material-related issues primarily concerning cables and wires. Despite having an abundance of cables, we often lacked the appropriate ones, complicating our efforts to devise a simple solution. After some experimentation, we managed to eliminate unnecessary cables and streamline the wiring system to a satisfactory level.

- ### Time Management

Time management was another significant challenge. Although each team member invested considerable individual effort into the project, our collective group work time was limited due to other commitments. Balancing these responsibilities with the project's demands was challenging, but we learned to manage our time more effectively as we progressed.